

Course Plan: Computer Networks**Credit: 3.00****Course Code: CSE 3103****Course Duration:** 16 Classes + 4 Class Tests + 4 Assignments**Target Audience:** Undergraduate students in Computer Science and Engineering

Chapter	Class	Topics Covered	
Introduction to Computer Networking	Class 1	Introduction to Networking Concepts - What is the Internet? - Protocols and network architecture - Network Edge and Core	
	Class 2	Understanding Network Performance and Security - Packet switching vs. Circuit switching - Delay, Loss, and Throughput in networks - Layered protocol models (OSI & TCP/IP)	
	Class 3	Class Test 1 (<i>Covers Class 1 & 2 topics</i>)	
Application Layer Functionalities	Class 4	Overview of Application Layer - Principles of network applications - Client-server vs. Peer-to-peer architecture	Assignment 1 (<i>Covers Class 1 & 2 & 3 topics</i>)
	Class 5	Web and HTTP - Web servers, HTTP/HTTPS protocol - Persistent vs. Non-persistent HTTP	
	Class 6	Email and Domain Name System (DNS) - SMTP, IMAP, and POP3 - DNS architecture, lookup mechanisms	
Transport Layer Functionalities	Class 7	Transport Layer Overview - Multiplexing and De-multiplexing - Transport services (TCP & UDP comparison)	Assignment 2 (<i>Covers Class 4 & 5 & 6 topics</i>)
	Class 8	TCP and UDP Protocols - TCP reliable transport, flow control - UDP functionalities, use cases	
	Class 9	Class Test 2 (<i>Covers Class 4 - 8 topics</i>)	
Network Layer Functionalities	Class 10	Data Plane - Forwarding and Routing - Inside a Router (input, output ports, switching fabric)	
	Class 11	Control Plane - Routing Algorithms (Link State, Distance Vector) - SDN and centralized control	
Data Link	Class 12	Data Link Layer Overview	Assignment 3

Layer & Wireless Networks		- Error detection and correction - Medium Access Control (MAC)	<i>(Covers Class 10 & 11 topics)</i>
	Class 13	Ethernet and VLANs - Ethernet frames, ARP - VLANs and Link Virtualization	
	Class 14	Wireless and Mobile Networks 802.11 (Wi-Fi) and Cellular Networks (4G, 5G) - Mobility management	
	Class 15	Class Test 3 <i>(Covers Class 10 - 14 topics)</i>	
Network Security	Class 16	Introduction to Network Security - Principles: Confidentiality, Integrity, Availability - Cryptographic techniques	
	Class 17	Security Mechanisms - Firewalls and Intrusion Detection Systems (IDS) - Secure Socket Layer (SSL) and IPsec	
	Class 18	Wireless and Transport Layer Security - TLS, Secure Email (PGP) - Security in 802.11 and mobile networks	Assignment 4 <i>(Covers Class 16 & 18 topics)</i>
	Class 19	Class Test 4 <i>(Covers Class 16 - 18 topics)</i>	
Course Wrap-up	Class 20	Emerging Trends in Networking SDN, IoT, Cloud Networking	
	Class 21	Final Review & Q&A Session	

Recommended Text Book:

- Computer Networking: A Top-Down Approach (8th Edition – Global Edition) - Jim Kurose, Keith Ross, Pearson Education Limited 2022.
- Data Communication and Networking (4th Edition) – Behrouz A. Forouzan (2006), Mc Graw-Hill, India.

Course Evaluation:

- Class Test – 10 (average of best 2)
- Assignment – 10 (average of best 2)
- Attendance – 8
- Semester Final – 72
- Total: 100**